

Report No.: TR-10C-0907-19-01-D

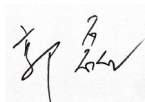
Safety test report

Product: Frequency Conversion Speed Regulation Three-Phase
Asynchronous Motor

Model: YZPBF400L3-10-200kW

Manufacturer: JIANGXI JIANGTE Electric Motor Co., Ltd

Safety test report

Sample name: Frequency Conversion Elevator Three-Phase Asynchronous Motor Model: YZPBF400L3-10-200kW Trademark:  Sample Number: 1 Serial: JC141420	Manufacturer: JIANGXI JIANGTE Electric Motor Co.Ltd Address: No.581, Yichun South Road, Yichun City, Jiangxi, P.R.China 
Test location: No.581, Yichun South Road, Yichun City, Jiangxi, P.R.China	
Test base: EN 60034-1:2010+AC:2010 Rotating electrical machines-Part 1 Rating and performance	
Test conclusion: This product has met test requirements of the standard for EN 60034-1:2004+AC:2010 Rotating electrical machines-Part 1 Rating and performance	
Tested by (name and signature): Zeng Shaojun 曾绍军 	Date 2021.03.22
Approved by (name and signature): Guo Lei 郭磊 	Date 2021.03.22
Onsite supervision (name and signature):	Date:
Remark: P = Pass, F = Fail, N = Not applicable in this test report	

Samples description

Equipment type: Movable ☐ Hand-held ☐ Transportable ☐ Stationary ☒
 Portable ☐ Direct plug-in ☐ Embed type ☐

Working mode: Continuous operation☒ short-time operation☐ intermittent operation☐

Power plug type: Single phase-two poles ☐ Single phase-three poles ☒
 Three phase-four poles ☐

Mode Connected to the supply: pluggable type A ☐ pluggable type B ☐
 Permanently connection ☒
 Detachable mains supply cord ☐ Non-detachable power supply cord ☐

Equipment classify: Class I ☒ class II ☐ class III ☐

Mass of equipment (kg):84

Outline size of equipment mm: $L_{max} \times B_{max} \times H_{max} = 2166 \times 900 \times 1400$

Safety interlock devices: Yes ☐ No ☒

IP degree: IPX0 ☐ IPX1 ☐ IPX2 ☐ IP23 ☐
 IP54 ☒ IP55 ☐ IPX6 ☐ IPX7 ☐

Over-current protection in primary circuits: Fuse ☐ Circuit-breaker ☐
 Type/specification: / Number : /
 Rating current: / Rating voltage: / locale: /

Structure form and fixed mode of supply coming conductors:
 Appliance coupler ☐ Other mode: fixed by bolt/nut ☒

Insulation between primary circuits and secondary circuits is depend on:
 Supply transformer (including switch type) ☐ opto-electricity coupler ☐
 Isolation transformer ☐ Isolation resistor ☐
 Electrical Clearance and creep distance ☐

Type of mains supply transformer: switch type transformer ☐
 Transformer for type silicon sheet steel core ☐

Connected to communicate network: Yes ☐ No ☒

Laser equipment: Yes ☐ No ☒ Laser equipment class: Laser wave-length:

Ionization radiation: Yes ☐ No ☒

Application of flame retardant material: Class V₁ Epoxy boards for Terminal board

Enclosure material: Metal

Nameplate/ drawing to samples

 通过ISO9001认证 ISO9001 Certified	型号Type YZPBF400L3-10			
	200kW	380V	IP 54	556r/min
	50 Hz	425A	S 1	3204 N.m
	1-50 Hz恒转矩 Constant-Torque	接法 Δ Connection		F 级绝缘 Insulation
	50-100 Hz恒功率 Constant-Power	40 ℃		
	Jm 23.1 kg.m ²	kg		
Q/JT023-2018	Jm 23.1 kg.m ²	Date 2021年1月		
销售电话 Tel 0795-3272270	JIANGXI JIANGTE Electric Motor Co., Ltd 江西江特电机有限公司			
FREQUENCY CONVERSION THREE-PHASE ASYNCHRONOUS MOTOR ADJUSTMENT SPEED FOR CRANE AND METALLURGICAL 起重及冶金用变频调速三相异步电动机				

Nameplate



Terminal box



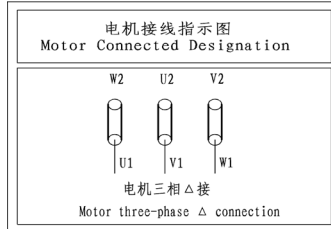
General View

1 Marks inspection

1.1 Test base: EN 60034-1, clause 10

1.2 Testing instrument: by inspection

Cl.	Requirement	Result	Verdict
10	TABLE: Rating plates		
10.2	Marking		
a)	The manufacturer's name or mark		P
b)	Manufacturer's serial number, or identification mark	JC141420	P
c)	Year of manufacture (or as code as part of item 2)	2021.1	
d)	Manufacturer's machine code	YZPBF400L3-10-200Kw	P
e)	For AC machines, number of phases	3~	P
f)	Number(s) of rating and performance standard(s) which are applicable (IEC 60034-X and/or equivalent national standard(s))	EN 60034-1	P
g)	Degree of protection provided by enclosures (IP code) in accordance with IEC 60034-5	IP54	P
h)	For motors within the scope of IEC 60034-30, the efficiency class (IE code) and the rated efficiency as specified in IEC 60034-30.	None	
i)	The thermal class and the limit of temperature or of temperature rise (when lower than that of the thermal class) and, if necessary, the method of measurement, followed in the case of a machine with a water-cooled heat exchanger by 'P' or 'S', depending on whether the temperature rise is measured above the primary or secondary coolant respectively (see 8.2). This information shall be given for both stator and rotor (separated by a slash) when their thermal class differ.	F	P
j)	Class(es) of rating of machine if designed for other than rating for continuous running duty type S6	S1	P
k)	Rated output(s) (W or VA)	200kW	P
l)	Rated voltage(s) or range of rated voltage (V)	300V	P
m)	For AC machines rated frequency or range of rated frequencies (Hz)	50Hz	P
	For universal motors, rated frequency (Hz) followed by appropriate symbol	None	N/A
n)	For synchronous machines excited by permanent magnets the open circuit voltage at rated speed.	None	N/A
o)	Rated current(s) (A)	428A	P
p)	Rated speed(s) or range of rated speeds (min ⁻¹ or 1/min)	596 min ⁻¹	P

q)	Permissible overspeed, if other than specified in 9.7 (min-1 or 1/min) Or the maximum safe operating speed if less than in 9.6 or if the machine is designed especially for variable speed operation.	Not other than specified in 9.7	N/A
r)	For DC machines with separate excitation or with shunt excitation and for synchronous machines, rated field voltage (V) and rated field current (A)	Not DC machines	P
s)	For AC machines, rated power factor(s)	0.758	
t)	For wound-motor induction machines rated open-circuit voltage (V) between slip-rings and rated slip-ring current (A)	Not wound-motor induction machines	N/A
u)	For DC motors with armatures intended to be supplied by static power converters, identification code of static power converter in accordance with IEC 60971 (alternatively for motors not exceeding 5 kW, rated form factor and rated alternating voltage at input terminals of static power converter, when this exceeds rated direct voltage of motor armature circuit)	Not DC motor	N/A
v)	Maximum permissible ambient temperature, if other than 40 °C; maximum permissible water temperature, if other than 25 °C (°C)	Not other than 40°C	P
w)	Minimum permissible ambient temperature if other than specified in 6.4 (°C)	Not other than specified in 6.4	N/A
x)	Altitude for which machine is designed (if exceeding 1 000 m above sea level)	Not exceeding 1000m	P
y)	For hydrogen-cooled machines, hydrogen pressure at rated output (Pa or bar)	Not hydrogen-cooled machines	N/A
z)	When specified, approximate total mass of machine, if exceeding 30 kg (kg)	3020kg	P
aa)	For machines suitable for operation in only one direction of rotation, direction of rotation, indicated by arrow; arrow easily visible	Reversible rotation	N/A
bb)	The connecting instructions in accordance with IEC 60034-8 by means of a diagram or text located near the terminals.		P
	Two different rated values shall be indicated by X/Y and a range of rated values shall be indicated by X–Y (see IEC 61293).	One rated value	P
	Except for normal maintenance, when a machine is repaired or refurbished an additional plate shall be provided to indicate the name of the company undertaking the work, the year of repair and the changes made.	new motor	-
Supplementary information:			

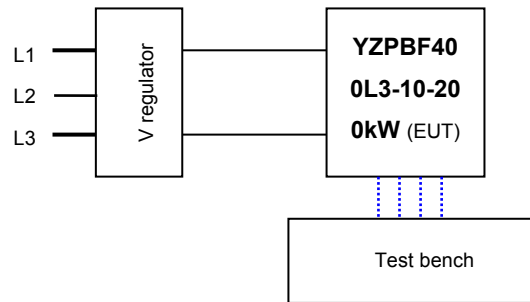
2 空载试验 motor of no-load test

2.1 Test base: EN 60034-1

2.2 Testing instrument: intermediate frequency generator set 中频发电机组, D26/1-W power meter (D26/1-W 功率表)、D26-V voltage meter (D26-V 电压表)、T19/1-A current meter (T19/1-A 电流表), MICR00HM METER(数字微欧计) PC9A

2.3 Test environment: Temperature: 16°C, Relative humidity: 49%

2.4 Test process:



- 1) Test set-up and connection is shown as above figure;
- 2) There are voltage meter, current meter and power meter on the test bench;
- 3) No-load current is measured by current meter and the Power is measured by power meter;
- 4) Stator resistance is measured immediately by MICR00HM METER after the test.

clause	Test description					verdict		
	空载试验 No-load test					P		
Operating condition	380V/50Hz							
stator resistor	R=0.00875Ω							
Note 1— No-load current I₀ 空载电流: I₀=(I₁+ I₂+ I₃)×倍率/3 Note 2—Input no-load power 输入空载功率: P₀= W₁± W₂ 功率×倍率 Note 3—No-load stator copper loss 空载定子铜损: P_{0CU1}=1.5I₀²R₁₀ Note 4—P₀′ =P_{Fe}+P_{fw}=P₀-P_{0CU1}								
No-load test original record 原始记录:								
No.	U0 (V)	I1 (A)	I2 (A)	I3 (A)	magnific ation	W1 (W)	W2 (W)	magnific ation
1	420.6	300.4	300.4	300.4	1	10230		1
2	380.91	213.76	213.76	213.76	1	6570		1
3	340.14	162.88	162.88	162.88	1	4640		1
4	300.12	135.51	135.51	135.51	1	3520		1
5	260.02	113.83	113.83	113.83	1	2660		1
6	220.31	94.84	94.84	94.84	1	1980		1
7	180.06	76.18	76.18	76.18	1	1510		1

Load test original record 原始记录:										
No.	U (V)	I1 (A)	I2 (A)	I3 (A)	magnifi cation	W1 (W)	W2 (W)	magnific ation	M (N.m)	N (r/min)
1	380	547.3	547.3	547.3	1	283420	0	1	4558	593.8
2	379.77	501.9	501.9	501.9	1	258500	0	1	4153	594.4
3	380.17	439.9	439.9	439.9	1	221190	0	1	3548	595.3
4	380.11	400	400	400	1	195370	0	1	3130.5	596
5	379.9	360.7	360.7	360.7	1	169150	0	1	2708	596.5
6	380.34	319.6	319.6	319.6	1	139140	0	1	2225	597.2

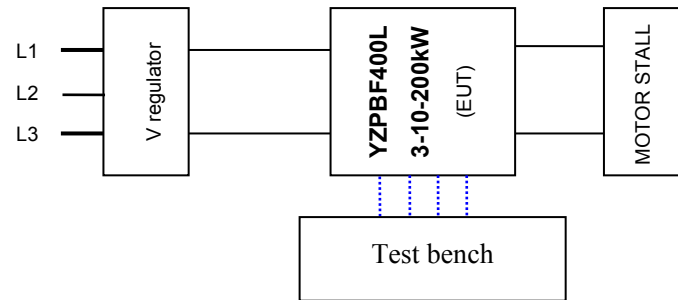
3 过电流试验 Occasional excess current test

3.1 Test base: EN 60034-1, Clause 9.3

3.2 Test instrument: Intermediate frequency generator set 中频发电机组, D26/1-W power meter (D26/1-W 功率表)、D26-V voltage meter (D26-V 电压表)、T19/1-A current meter (T19/1-A 电流表),

3.3 Test environment: Temperature:16°C, Relative humidity: 49%

3.4 Test process:



- 1) Test set-up and connection is shown as above figure
- 2) Before the test the motor should be locked.
- 3) There are voltage meter, current meter and power meter on the test bench.
- 4) Occasional excess current is measured by current meter and the Power is measured by power meter.

Clause		Test description								verdict	
9.3		Occasional excess current								P	
Operating condition		50Hz,△接法									
Test voltage (V)	pole	电流 (A) Current (A)					功率 (W) Power (W)			Test Duration	Test result
		I _{AB}	I _{BC}	I _{CA}	Multiple	I _K (A) (Note 1)	P _A ±P _C	Multiple	P _K (W) (Note 2)		
380	10	684.9	684.9	684.9	1	684.9		1	320	1min	No breakdown

Note 1—locked rotor current (堵转电流) $I_K = (I_1 + I_2 + I_3) \times \text{Multiple} / 3$;
Note 2—locked rotor power (堵转功率) $P_K = P_A \pm P_C \times \text{Multiple}$
Note 3— the required current equal to 1.5 time the rated current, the rated current is 428.1A.

Locked test original record 1 原始记录									
No.	U _k (V)	I ₁ (A)	I ₂ (A)	I ₃ (A)	Magnification	W ₁ (W)	W ₂ (W)	Magnification	T _k (Nm)
1	279.53	1928.1	1928.1	1928.1	1	221300	0	1	2704.93
2	255.54	1716.2	1716.2	1716.2	1	175880	0	1	2151.42
3	222.26	1431	1431	1431	1	122930	0	1	1505.66
4	190.85	1174.8	1174.8	1174.8	1	83080	0	1	1018.25
5	165.65	980.7	980.7	980.7	1	58090	0	1	712.55
6	135.29	763.3	763.3	763.3	1	35380	0	1	434.54

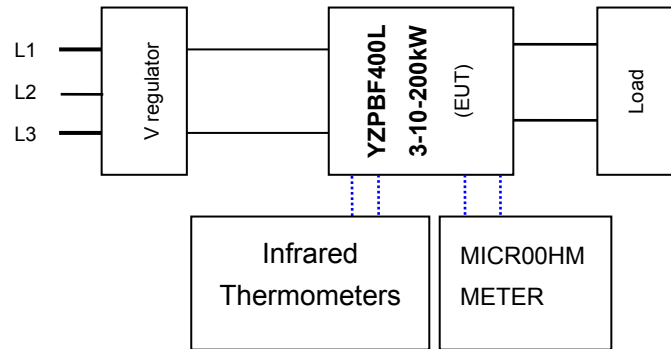
4 温升试验 Temperature Rise test

4.1 Test base: EN 60034-1, Clause 8

4.2 Test instrument: PT40 Infrared Thermometer(红外线测温仪), 直流电动机 Z2-72, 三相异步电动机 JW5624, 转速转矩传感器 1C1A, MICR00HM METER(数字微欧计) PC9A

4.3 Test environment: Temperature:16℃, Relative humidity:48%

4.4 Test process:



1) Test set-up and connection is shown as above figure;

2) Before the test, measure the convenient value of coolant temperature winding resistance R_1 by MICR00HM METER. The coolant temperature θ_1 is the average temperature of different points;

(试验前测量冷态冷却介质温度 θ_1 和绕组电阻 R_1 , 介质温度是不同测量点测得的温度值的平均值。)

3) The value to be adopted for the temperature θ_2 of a coolant during a test shall be the mean of the readings of the temperature detectors taken at equal intervals of time during the last quarter of the duration of the test;

(热态冷却介质温度 θ_2 是在整个试验期间的最后 1/4 时间内, 按相等时间间隔测得的几个温度计读数的平均值, 用 TP9016U 多路温度记录仪测量。)

4) After the test, measured the temperature of the terminal, Bearing, iron core by infrared thermometer;

(试验结束时, 测量端子、轴承、铁芯的温度, 用 PT40 红外线测温仪测量。)

5) After the test, the initial resistance reading is obtained within 90s;

(试验结束后, 在 90 秒内测量完绕组电阻值 R_2)

6) The final wind temperature θ is calculated by formula: $\theta = R_2/R_1 (k + \theta_1) - k$;

7) The temperature rise θ_r is calculated by formula: $\theta_r = \theta - \theta_2$;

Clause		Test description			verdict
8.3.3		Temperature of coolant			P
Operating condition		Before temperature rise test			
Pole	ambient temperature θ_1 (°C)	Temp. terminal θ_2 '(°C)	Requirement θ_1 - θ_2 '	Meas. θ_1 - θ_2 '	
10	16	17	No exceed 2K	1	

Clause			Test description									verdict	
8.3.4			Measurement of coolant temperature during test										
Testing time			During the last quarter of the duration of the test.(整个试验期间的最后 1/4 时间)										
Time	U V	I1 A	I2 A	I3 A	magnifyi ng power	W1 W	W2 W	M N.m	N r/min	P kW	T _{IC} ℃	T _{AT} ℃	
13: 20	380	399.1	399.1	399.1	1	211740	0	3204	596	200	16	52.2	
13: 50	379	400	400	400	1	211350	0	3204	596	200	16	52.5	
14: 20	378	398	398	398	1	210570	0	3204	596	200	17	52.7	
14: 50	381	397.3	397.3	397.3	1	211380	0	3204	596	200	17	52.9	
15: 20	381	402.9	402.9	402.9	1	214240	0	3204	596	200	17	53.1	
15: 50	378	400.9	400.9	400.9	1	212360	0	3204	596	200	17	53.3	
After testing													
T(s)	39		47		52		57		64		69		
R(Ω)	0.00975		0.00974		0.00973		0.009725		0.009715		0.00971		
Resistance method Temperature Measurements (电阻法测温升)													
θ ₁ (℃)		R (Ω)				k			θ ₂ (℃)		θ (℃)		
16		0.0098				235			17		86.75		
Tr temperature rise				Tr max limit					Verdict				
69.75				105					P				
Note 1 – R ₁ = initial resistance, R ₂ = final resistance; Note 2 – θ=final winding temperature, θ=R ₂ /R ₁ (k+θ ₁)-k; Note 3 – tr = temperature rise, tr=θ-θ ₂ ; Note 4 –T _c = tr corrected (tc = tr – {ta2 – ta1}+[40℃ or max. rated ambient]); Note 5 –Trmax = maximum permitted temperature, table 7 in EN 60034-1 Note 6 –For copper k = 235, for aluminium k = 225.													

Ch. No.	Location / Part (by thermocouple)	Temperature (°C)				Limit (°C)
1	Terminal body (端子温度)	54	54	54	54.6	105
2	Motor lead wire (引线温度)	40	40	40	40	105
3	Measurement of bearing temperature (Clause 8.9)	60	62	61	60	135
5	Motor body (External surface)	53.3	53.4	53.4	53.5	105

5 耐电压试验 Withstand voltage test**5.1 Test base:** EN 60034-1, Clause 9.2**5.2 Test instrument:** Dielectric strength test meter, IG5-0.5**5.3 Test environment:** Temperature: 16℃, Relative humidity: 46%**5.4 Test process:**

Clause			Test description			Verdict
9.2			withstand voltage test (耐电压试验)			P
Test operating conditions:			Go on the withstand voltage test at once after the heating test; 50Hz; Sine wave (热试验后马上进行耐电压试验; 50Hz, 正弦波)			
Test position	Rated voltage U_N	pole	test require voltage $2U_N+1000V$	Applied voltage U_t	Test Duration	Test result
L1,2,3 to GND	380V	10	1760V	1760V	1 min	No breakdown

6 振动试验 vibration test

6.1 Test base: EN 60034-14

6.2 Test instrument: Pocketable vibration meter, Vm-63a

6.3 Test environment: Temperature: 16℃, Relative humidity: 48%

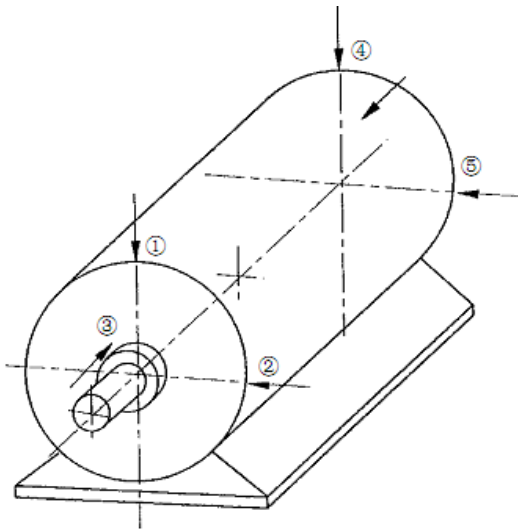
6.4 Test process:

Clause		Test description						Verdict	
		Vibration test （振动试验）						P	
Operating conditions:		380V/50Hz							
test point （测试点）		①	②	③	④	⑤	⑥	Max.	Limit value Note 2
A-Weighted（计权） Number（编号）		mm/s	mm/s	mm/s	mm/s	mm/s	mm/s	mm/s	mm/s
pole	10	0.7	0.5	0.6	0.8	0.7	0.6	0.8	1.8

Note 1—The maximum value of the points readings regard as the measured value of the tested motor.

(测定结果以各测点读数的最大值作为被测电机的实测值)

Note 2 free suspension 弹性安装



Note — point ⑥ is the opposite of point ③.

7 noise test 噪声试验**7.1 Test base:** EN 60034-9**7.2 Test instrument/equipment:** HS5633B General acoustimeter (通用声级仪)**7.3 Test environment:** Temperature: 16°C, Relative humidity: 48%**7.4 Test process:**

Clause		Test description						Verdict	
		Noise test (噪声试验)						P	
Operating conditions:		380V/50Hz，Background noise（环境噪声）48dB							
test point（测试点）		L1 (front)	L2 (back)	L3 (left)	L4 (right)	L5 (above)	SPL Lp 平均声压级	sound power level Lw 声功率级	sound power level criterion limit
A-Weighted(计权) Number（编号）		dB	dB	dB	dB	dB	dB	dB	dB
pole	10	85	87	85	84	88	85.8	93.8	99
Note 1 — dB-SPL（平均声压级） $L_P = (L_1 + L_2 + L_3 + L_4 + L_5)/5$;									
Note 2—sound power level（声功率级） $L_W = L_P + 10\lg(S/S_0)$, which $S = 2\pi r^2$, S_0 is surface basis, $S_0 = 1\text{m}^2$, r is radius measurable（测量半径）, $r = 1\text{m} + 0.5$ time maximum length of the motor size（1m+0.5 倍的电机最大长度的尺寸。电机最大长度的尺寸）									

8 clearance and creepage distance measurements (爬电距离和电气间隙)**8.1 Test base:** EN 60664-1**8.2 Test instrument:** Vernier Caliper**8.3 Test environment:** Temperature: 16°C, Relative humidity: 50%**8.4 Test process:**

Creepage distance, clearances and distance through insulation are measured by Vernier Caliper

Clause	Test description					Verdict
	TABLE HD 625.1 S1 / IEC 60664-1: clearance and creepage distance measurements					P
	U r.m.s. (V)	Required cl (mm)	Meas. cl (mm)	required dcr (mm)	Meas. dcr (mm)	Verdict
Basic insulation						
Winding – stator	380	10	10	10	10	Ok
Winding – rotor	380	10	10	10	10	Ok
Winding – frame	380	10	10	10	10	Ok
Line conductor – frame	380	10	10	10	10	Ok
Functional insulation						Ok
Winding – Winding	380	10	10	10	10	Ok
Terminal – Terminal	380	10	10	10	10	Ok
Note 1- Requirement for clearance exceeds requirement for creepage distance.						
The requirements in this table is used as reference in the test						

9 Overspeed 超速试验

- 9.1 Test base: EN 60034-1, Clause 9.6, Clause 9.7
- 9.2 Test instrument: VARISPEED 616G5, CIMR-G5A4037; handy digital tachometer(手持数字转速表), SZG-441A.
- 9.3 Test environment: Temperature: 16℃, Relative humidity: 48%
- 9.4 Test process:

Clause		Test description			Verdict
9.7		Overspeed test(超速试验)			P
Operating conditions:		690V, sine wave			
Testing instrument					
pole	Rated speed r/min	Required Overspeed r/min	Test Overspeed values r/min	Test duration 试验历时时间	Test result
10	593	770	770	2 min	No breakdown
a- safe operating speed of cage induction motors, table 17. b—1.2 time for rated speed Note — synchronous speed =60f/r					

10 Momentary excess torques for test**10.1 Test base:** EN 60034-1, Clause 9.4**10.2 Test instrument:** 1C1A revolving speed Torque Transducers (转速转矩传感器), JW-2A torque gauge (扭矩仪)**10.3 Test environment:** Temperature: 16℃, Relative humidity: 48%**10.4 Test process:**

Clause		Test description			Verdict		
9.4		Momentary excess torque test 超转矩试验			P		
Operating conditions		50Hz/380V, sine wave					
pole	Rated torque N·m	Required Excess torque N·m (Note 1)	Excess torque value N·m (过转矩值)	Test duration 试验历行时间	Test result		
10	3204	5126	5126	15s	No breakdown		
Note 1—at least 1.6 times rated torque							
Records:							
U (V)	I ₁ (A)	I ₂ (A)	I ₃ (A)	Multiple	W ₁ (kW)	W ₂ (W)	Multiple
380	684.9	684.9	684.9	1.0	320	0	1.0

11 Safety test

10.1 Test base: EN 60204-11,

10.2 Test instrument: Standard test Finger

10.3 Test environment: Temperature: 16°C, Relative humidity: 48%

10.4 Test process:

Clause		Test description	Verdict	
EN 60204-11 Accessibility		test Finger shall not touch the moving part and Live part	P	
pole	Accessibility		Remark	Test result
1	Can touch the Moving part		N	P
2	Can touch the Live Part?		N	P